Martinez, Thomas - Dissertation Abstract

We study affine patches of positroid varieties through periodic combinatorial models. Open positroid varieties are subvarieties of Grassmannians specified by rank conditions on cyclically consecutive columns of matrix representatives. Requiring a Plücker coordinate to be nonzero gives an affine patch, which can be identified with an open Richardson variety in the affine flag variety. These identifications bring the combinatorics of affine permutations to bear on the geometry of positroid varieties. Our central aim is to develop explicit diagrammatic tools for understanding the decompositions, enumerative invariants, and coordinate structures of these patches.

We introduce affine Deograms, periodic strand diagrams that encode the affine Deodhar decomposition of a patch into products of algebraic tori and affine spaces. These diagrams make the individual pieces and their dimensions combinatorially accessible. Local transformations relate the diagrams associated with different Plücker charts, providing a common framework for studying the patches and extracting combinatorial information from their geometry.

Using this framework, we characterize when an affine patch is itself an algebraic torus. We prove that this occurs precisely when its Deodhar decomposition has a single piece, equivalently when the corresponding Bruhat interval avoids a specified six-element subinterval called a 2-crown, or when its finite-field point-count polynomial is a pure power of $q-1$. Complementing this classification, we connect the standard diagonal torus action to link topology. For type A braid Richardson varieties, the existence of a dense torus orbit is equivalent to smooth sliceness of the associated link; for open positroid varieties, it is also equivalent to that link being an unlink.

The combinatorics of these decompositions also yields new connections with rational Catalan theory. We construct a recursive bijection between maximal Deograms and rational Dyck paths for coprime parameters, using diagram transformations to match geometric reductions with decompositions of lattice paths. In the Fuss–Catalan case, we give explicit nonrecursive bijections, including a correspondence between twisted maximal Deograms and rational parking functions.

Finally, we develop diagrammatic models for cluster coordinates on affine patches. We introduce affine three-dimensional plabic graphs, periodic bicolored graphs whose local transformations encode mutations and compatible rescalings of candidate cluster seeds. We prove that the associated exchange matrices are really full rank and that the resulting cluster algebras are locally acyclic, hence equal to their upper cluster algebras. For positroid patches, these graphs admit a description by punctured plabic graphs, whose move-equivalence classes correspond to affine Le-diagrams. These constructions provide candidate cluster structures on the patches and extend to the broader setting of double affine braid varieties.